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CLINICAL AND LABORATORY NOTES

MENTAL ARITHMETIC AND EYE OPENING. AN EEG FREQUENCY ANALYSIS AND GSR STUDY

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INTRODUCTION

The EEG changes occurring when the subject opened his eyes or when he was solving a difficult problem have been extensively studied since the early thirties. First of all, similarities between the EEG effects of these conditions were observed. As the motivational state of the subjects began to be studied more closely, it was noticed that the difference between them might be more interesting.

The present experiment was intended as an investigation of the differences between the simple opening of the eyes on the one hand and two problem-solving situations at different levels of general activation on the other hand. Simultaneously it was hoped that EEG might help to discriminate between those two latter situations. The galvanic skin response (GSR) was used as a monitor of the level of general activation.

MATERIAL AND METHODS

Our sample consisted of twelve healthy, paid volunteers (eleven men and a woman) between 19 and 31 years of age; mean age 25.8 years.

The experiment proper was preceded by a conventional EEG recording of all subjects. EEG and GSR were taken during the experiment. Two pairs of GSR electrodes were used; the first pair on the second finger and the second pair on the fourth finger of the left hand. Changes of skin resistance were measured. An EEG of the right parieto-temporal region (electrodes P4-T6) was tape-recorded (equipment described by Mikiska 1962) and subsequently subjected to frequency analysis. Our 16-channel electroencephalograph, the DC polygraph and the 1-channel analyser were manufactured by Kaiser Co. The analyser has 6 frequency bands: delta (1.5–3.5 c/sec), theta (3.5 –7.5 c/sec), alpha 1 (7.5–9.5 c/sec), alpha 2 (9.5–12.5 c/sec), beta 1 (12.5–17.5 c/sec), beta 2 (17.5–25 c/sec). The broad-band filters were stable, no tuning was necessary. The readout circuits were calibrated daily by means of sine signals of constant voltage, so that each of the output pulses of the integrators represented 1 μV-sec. For a more detailed description of the analyser see Kaiser *et al.* (1964).

EXPERIMENTAL PROCEDURE

Each subject was examined in the following conditions: C: eyes closed; O: eyes opened; A: eyes closed, mental arithmetic; M: eyes closed, mental arithmetic with financial reward. Two sequences of conditions were used: C-O-A-M and C-O-M-A. Half of the subjects experienced each sequence.

Mental arithmetic consisted in serial subtraction; the second problem was made somewhat more difficult than the first in order to compensate for the effect of training. Responses to problems were given orally immediately after the end of each condition. The sequence of problems remained constant. Immediately prior to condition M the subjects were told that the amount of reward was determined by the speed and accuracy of the solution. On the other hand, immediately prior to condition A the subjects were informed that their performance would be observed, however, without any effect on the amount of reward.

Data evaluation

Each of the experimental conditions averaged 45–60 sec. Subsequently a part of the tracing, totalling 45 sec of artifact-free recordings, was selected from each condition for frequency analysis. This part was divided into 9 epochs of 5 sec each. Most of the epochs were contiguous. The number of pulses of the integrators during the 9 epochs served as basic data for statistical calculations.

The number of GSRs was determined for each particular condition from the corresponding 9 epochs of the original recording. The following were the criteria for a scorable GSR: (a) A definable beginning and end. We did not score gradual changes of more than 10 sec duration, or indefinite fluctuations in which the gradient angle was less than 5° (*i.e.* a minimum of a 1 % change of the original level of resistance per sec. (b) Some part of the GSR must take place within an epoch or within a second after the termination of an epoch. (c) The GSR must be manifest in both derivations used.

RESULTS

Visual evaluation of the routine EEG revealed normal curves in ten subjects; the remaining two recordings were described as borderline.

For a summary of the number of GSRs, the reader is referred to Table I. Evaluation of the number of GSRs (by means of the sign test) did not reveal any significant differences between conditions C and O, C and A, O and

A. Significantly more GSRs occurred in condition M than in any other condition ($P < 0.01$). Moreover, in any given subject the number of GSRs in conditions C, O and A never exceeded the number registered in condition M.

TABLE I

Number of GSRs, sequence of conditions

Name	No. of GSRs				Sequence
	C	O	A	M	
M.	2	7	9	9	M–A
O.	0	0	4	5	M–A
D.	0	0	0	0	M–A
B.	0	0	2	4	M–A
K.	0	0	0	7	M–A
V.	0	0	5	7	M–A
Z.	1	0	2	2	A–M
A.	0	0	7	8	A–M
I.	3	4	1	5	A–M
P.	0	0	0	2	A–M
C.	0	0	0	0	A–M
H.	1	2	1	1	A–M
Total	7	13	31	50	

EEG frequency analysis

1. The differences between experimental conditions in the quantity (abundance) of particular frequency bands are given in Fig. 1. The quantity was ascertained by adding the number of pulses produced by the integrators over the given time span (45 sec). Output values of the integrators for conditions O, A and M are given in proportion to the values for condition C (relative quantity). We used this method, *inter alia*, for the purpose of reducing the effect of differences in the initial EEG level.

It follows from Fig. 1 that: (a) in condition O relative quantities of all frequency bands were lower than in conditions A and M; (b) in conditions O, A and M there was less theta and alpha activity than in condition C; (c) while suppressed in condition O, beta activity increased in conditions A and M; (d) differences between conditions

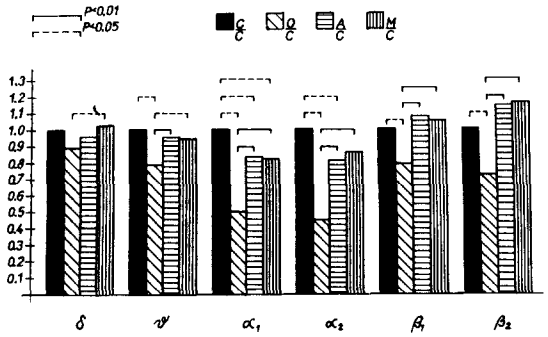

Fig. 1

The quantities of activity in various conditions standardized by condition C. The bridges above indicate the conditions which differ significantly (t test for matched pairs).

A and M were not significant.

2. We were also interested in the possible effect of experimental conditions on EEG variability in time. The data (integrator output values for 5 sec epochs) were subjected to logarithmic conversion. Standard deviations of the converted data were computed for each subject, frequency band and experimental condition. These standard deviations can be considered to be closely related to the coefficient of variance. As shown by Fig. 2 mental activity is accompanied by decreased variability of beta activity.

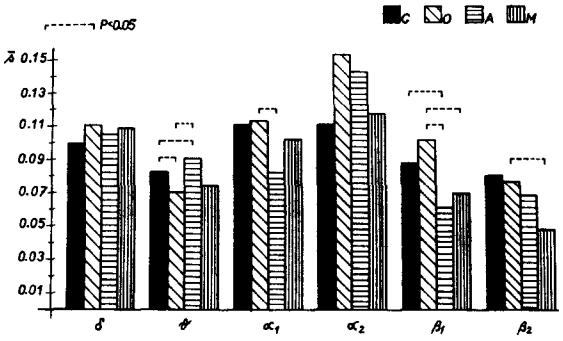

Fig. 2

Variability of the output of the integrators during different experimental conditions. The value $\bar{s}$ on the vertical axis is a mean standard deviation of the logarithmically converted outputs produced in 9 consecutive epochs (5 sec each). The bridges above indicate the conditions which differ significantly (t test for matched pairs).

DISCUSSION

Eye opening as well as mental arithmetic suppressed the theta and alpha quantities; the difference between the two conditions consisted just in the extent of suppression. It was beta activity which was found to behave in opposite ways, decreasing with eyes open and increasing with mental arithmetic. This was the most conspicuous difference between the two conditions.

Suppression of fast rhythms by opening the eyes is in disagreement with some generally established views; however, it conforms with the results of Mundy-Castle (1957), Shminke (1961) and Motokizawa and Fujimori (1964). Increase in beta activity in connection with mental arithmetic is in agreement with the observation of Mundy-Castle (1957). We excluded muscle artifacts by careful observation of the primary curves.

A consistent decrease in EEG variability was obtained in beta bands whose quantity was demonstrated to increase with mental activity. Some of our previous observations indicate that EEG variability rises with fluctuations of the level of vigilance; the decreased beta variability might be explained by the steady level of vigilance accompanying heightened attentiveness.

We have demonstrated that the reward condition (M) was effective enough to raise the level of general activation as measured by the number of GSRs. However, the fre-

quency analysis used was unable to discriminate between the rewarded and non-rewarded mental arithmetic conditions. This failure can hardly be ascribed to a low reliability of the analyser; the average levels of both mentioned conditions are practically identical in all frequency bands under study.

It might be possible that the broad-band analysis we used was not sufficiently sensitive. Glass (1964) was able to demonstrate some EEG changes due to an increased motivational state in the subjects performing mental arithmetic. He believed that the changes consisted in a more intense alpha blocking. However, a detailed study of his method evokes some doubts about the nature of the ascertained changes. He measured the length of his ink traces by means of an opisometer, failing, however, to state any frequency limitations for his measurements. The length of a trace can of course be decreased not only by a decrease of amplitude (this would be indicative of "blocking"), but by a decrease of frequency as well.

The EEG changes at different levels of general activation still remain largely unexplored, providing an interesting field for further studies.

SUMMARY

1. Twelve healthy subjects were examined. EEG and GSR were taken when the subjects were resting with their eyes closed (C), resting with their eyes open (O), performing mental arithmetic with their eyes closed (A), and performing mental arithmetic with their eyes closed and promised financial reward (M).

2. The number of GSRs in condition M was significantly higher than in any other condition.

3. Frequency analysis was made of the EEG of the right parieto-temporal region and the number of GSRs was ascertained according to precisely specified criteria.

4. In condition O relative quantities were on the average lower in all frequency bands than in conditions A and M.

5. In conditions O, A and M theta and alpha activities registered a lower level than in condition C.

6. In condition O beta activity decreased, while increasing on the average in conditions A and M.

7. Compared with conditions C and O, the variability of beta activity in conditions M and A was lower.

8. Frequency analysis did not disclose any significant difference between conditions A and M.

9. The results indicate that beta activity (both its amount and variability) is more closely related to the general level of activation than are the brain waves in any other frequency band under study.

RÉSUMÉ

CALCUL MENTAL ET OUVERTURE DES YEUX. ANALYSE DE FRÉQUENCE EEG ET ÉTUDE ÉLECTRODERMOGRAPHIQUE

1. Douze sujets bien portants ont été examinés. L'EEG et l'électrodermogramme ont été enregistrés à l'état de repos, yeux fermés (C), à l'état de repos, yeux ouverts (O), pendant une épreuve de calcul mental, les yeux fermés (A), et pendant une épreuve de calcul mental, yeux fermés, avec promesse de récompense financière (M).

2. Le nombre d'électrodermogrammes (EDG) en condition M est de manière significative plus élevé qu'au cours de toute autre condition.

3. L'analyse de fréquence a été faite sur l'EEG de la région temporo-pariétale droite et le nombre des EDGs a été vérifié suivant des critères décrits de façon précise.

4. En condition O, les quantités relatives de fréquences sont en moyenne plus faibles dans toutes les bandes de fréquences qu'en condition A et M.

5. En conditions O, A et M, les activités theta et alpha sont moindres qu'en condition C.

6. En condition O, l'activité beta diminue, tandis qu'elle augmente en moyenne en conditions A et M.

7. Par comparaison aux conditions C et O, la variabilité de l'activité beta est moindre en conditions M et A.

8. L'analyse de fréquence n'a mis en évidence aucune différence significative entre les conditions A et M.

9. Les résultats indiquent que l'activité beta (à la fois sa quantité et sa variabilité) est en relation plus étroite avec le niveau général d'activation que ne le sont les autres ondes cérébrales dans toute autre gamme de fréquence étudiée.

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